# Performance Testing

| | |
|---|---|
| **Date** | 1 October 2026 |
| **Team ID** | 6ab63662000bf9bb96385cb4 |
| **Project Name** | Auto Ticket Classification using Flow Designer |
| **Maximum Marks** | 3 Marks |

## Performance Testing

This document captures the performance testing strategy, test cases, and results for the Auto Ticket Classification system.

## 1. Testing Strategy

Testing was conducted using ServiceNow's Automated Test Framework (ATF) for functional tests and manual load simulation for performance benchmarks. The testing environment was a ServiceNow sub-production (test) instance mirroring production configuration.

## 2. Test Cases & Results

| Test ID | Test Case | Type | Input / Scenario | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC-01 | Single ticket auto-classification | Functional | Create 1 incident with clear short description | Category, Subcategory, Assignment Group auto-populated within 3s | Fields populated in 1.8s with 92% confidence | Pass |
| TC-02 | Low confidence handling | Functional | Create incident with vague/ambiguous description | Ticket routed to manual triage queue, work note added | Correctly flagged (confidence: 54%), routed to triage | Pass |
| TC-03 | Manual override | Functional | Agent changes auto-assigned category | Override logged, original prediction stored for retraining | Override captured in audit log correctly | Pass |
| TC-04 | Bulk ticket classification | Performance | Create 50 incidents in rapid succession | All 50 classified within 2 minutes total | All 50 classified in 1 min 42 sec | Pass |
| TC-05 | Classification accuracy | Accuracy | Run model against 500 test tickets with known categories | Accuracy >= 85% | Accuracy: 88.4% (442/500 correct) | Pass |

| | | | | | | |
|---|---|---|---|---|---|---|
| TC-06 | Email-created incident | Integration | Send incident via inbound email | Incident created and auto-classified | Classified correctly with 86% confidence | Pass |
| TC-07 | PI model retraining | Functional | Trigger scheduled retraining job | Model retrained without errors, accuracy maintained | Retrained successfully; accuracy: 89.1% | Pass |
| TC-08 | Dashboard data accuracy | Functional | Verify PA dashboard metrics match actual data | All KPIs reflect correct values | All metrics verified correct | Pass |

## 3. Performance Metrics Summary

| Metric | Target | Achieved | Status |
|---|---|---|---|
| Classification Response Time | < 3 seconds | 1.8 seconds (avg) | Met |
| ML Model Accuracy | >= 85% | 88.4% | Met |
| Throughput (tickets/hour) | >= 500 | ~1,760 tickets/hour | Exceeded |
| System Availability | 99.9% | 99.95% (during test period) | Met |
| Misrouting Rate | < 5% | 3.2% | Met |
| SLA Compliance Improvement | >= 20% | 32% improvement | Exceeded |